Skills and Comptencies for the Sustainability Engineer

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1.1 Difference between Skills and Competences

The difference between skills and competences is that a skill is the ability to perform a particular task, whereas competence is the broader ability to apply skills, knowledge, and judgement effectively in a given situation.

Skills: A is an ability that you develop through learning, practice, or experience. It usually relates to performing a specific task.

Competence: A competence is the ability to use your **knowledge** and **skills** appropriately to handle a task, solve a problem, or fulfil a responsibility. It can also involve attitudes, communication and judgement.

Normative, Strategic and Systems thinking are the three major competences required for the sustainability Engineer.

Questions

1. What will the engineer for tomorrow look like?
2. How will they be trained?
3. What competences will thy need?

2 UNESCO's Key Competences for sustainability

1. Systems-thinking competency.
2. Anticipatory Competency.
3. Normative Competence
4. Strategic Competency.
5. Collaboration Competency.
6. Critical-thinking competency.
7. Self-awareness competency.
8. Integrated problem-solving competency.

References

[1] U. Beagon et al. “Preparing engineering students for the challenges of the SDGs: what competences are required?” In: *European Journal of Engineering Education* 48.1 (2023), pp. 1–23.